

A Classifier for Spinal Tumors using Novel Data Augmentation

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Abstract. Detection and classification of spinal tumors remain a challenge due to the lack of available annotated data. This motivated us to create an augmented database for researchers to use. This study introduces a novel data augmentation technique in which we insert tumors along the cerebrospinal fluid in healthy lumbar spine MRI scans. This method addresses the scarcity of labeled data by generating anatomically realistic tumor variations, significantly enhancing training diversity and model robustness. We have systematically generated the augmented data with 100% accuracy in augmentation. Using this dataset, a fully automated pipeline achieves 99% accuracy in classifying spinal tumor types from T2-weighted MRI images. The system has a 6-layer Convolutional Neural Network (CNN) strategy and incorporates anatomical priors for context-aware predictions. This augmentation-driven approach offers a clinically relevant solution for spinal tumor diagnosis.

Keywords: Data Augmentation · Classification · Random Forest · CNN

1 Introduction

Spinal tumors—whether intradural or extradural—pose critical diagnostic challenges due to their varying morphology, subtle appearance in MRI scans, and anatomical proximity to critical spinal structures. These tumors are typically classified into three major categories based on their anatomical location: *intradural* (within the spinal cord), *intradural extramedullary* (within the dura but outside the spinal cord), and *extradural* (outside the dura), as depicted in Fig. 1. Accurate classification of these tumor types is essential for appropriate clinical management and surgical planning.

Despite significant advances in computer-aided diagnosis, there exists a notable gap in automated systems designed specifically for spinal tumor classification. Most existing models focus on other anatomical regions, while spinal

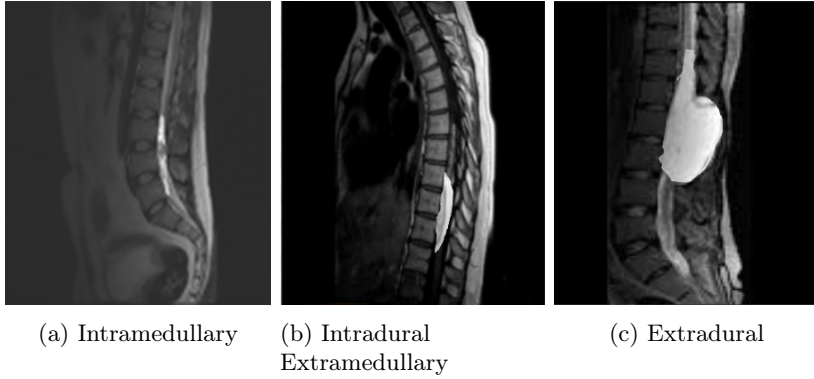


Fig. 1: Types of spine tumor.

tumor classification remains underexplored, partly due to the limited availability of labeled data and the complexity of spinal anatomy.

In this study, we present a spine tumor data set along with a fully automated pipeline that performs classification of spinal tumor types from T2-weighted lumbar spine MRI images with a high degree of accuracy (**99%**) [5]. Our approach is grounded in two key innovations: domain-specific **tumor data augmentation** and a classifier with **Convolutional Neural Network (CNN)** models.

The main contributions of this work are:

- **Data augmentation via synthetic tumor insertion:** Given the scarcity of labeled synthetically insert tumor-like regions along the *cerebrospinal fluid (CSF)* in healthy lumbar spine MRI scans to expand the training dataset significantly, introduce anatomical variability, and enhance the robustness of the classification model. We have achieved 100% accuracy in data augmentation with no garbage data generation.
- **Classification approach:** A 6-layered CNN is used to classify the tumors into one of the three anatomical types based on localized image patches. This combination leverages the interpretability of classical methods and the power of feature learning capabilities in deep learning.

By addressing the challenges of limited annotated data and complex tumor localization, our approach provides a robust, clinically relevant solution for spinal tumor classification. It serves as a step toward more reliable computer-aided diagnosis tools for spinal oncology and can significantly assist radiologists in decision-making processes.

2 Related Works

Due to the scarcity of annotated clinical MRI data, a handful of augmentation techniques have been employed, such as Mixup [24], GAN [11], CycleGAN [25], and CutMix [22]. CutMix developed a regularization technique in the field of data augmentation for deep learning. This technique creates training samples by cutting and pasting patches among training images while proportionally merging their labels. This improves localization and generalization by pushing the model to concentrate on less discriminative areas. Similarly, to promote smoother decision limits and enhanced robustness to adversarial cases, Mixup suggested a linear interpolation technique between random pairings of training examples and their labels. The underlying premise of CutMix and Mixup, however, is that semantic information can be maintained via straightforward spatial or pixel-level interpolations; this may not be the case in situations with intricate object arrangements or substantial occlusions. By employing cycle-consistent adversarial networks to tackle the problem of unpaired image-to-image translation, CycleGAN made realistic domain transfers possible without the need for paired training data. Despite its strength, CycleGAN’s augmentation frequently introduces artifacts and lacks control over semantic accuracy, which makes it less dependable for jobs requiring the preservation of fine-grained features. These augmentation techniques can suffer from problems like semantic ambiguity, label mixing artifacts, or uncontrolled changes that could reduce model reliability in safety-critical applications, even though they generally greatly improve performance in many tasks. Among all these existing techniques, GANs are widely used for generating realistic, anatomically plausible MRI slices in low-data regimes, enhancing classifier generalization. However, GANs often suffer from mode collapse and training instability, especially in spine MRI where tumors are small

and embedded in complex, low-contrast backgrounds. These limitations lead to artifacts, blurry outputs, and biased generation due to class imbalance. The main problem with all the existing techniques is that they are not domain specific and leads to a lot of garbage data generation which cannot actually be used in the training, thus the task of assorting the relevant data becomes an added burden. In this paper, we have tried to propose an augmentation technique which will work specifically for spine tumors in the cerebrospinal fluids and all the data generated through our augmentation technique can be used for training using any deep learning technique.

While extensive literature exists for brain tumor classification, spine tumor classification remains underexplored. Havaei et al.[10] and Chen et al.[3] introduced cascaded CNNs combining local and global features, though lacking modern depth. Dong et al.[6] used U-Net for segmentation under limited data but faced memory overhead. Bernal et al.[2] offered a foundational CNN review, albeit outdated. Hybrid approaches like those by Fernando & Tsokos [7] and AlTahhan et al.[1] enhance interpretability and accuracy but may compromise generalization. Talo et al. [21] achieved high accuracy with ResNet34, risking overfitting, while Kumar et al. [15] effectively employed transfer learning for imbalanced datasets but rely heavily on pretrained models. These methods, though effective for brain MRIs, do not generalize well to spine tumors due to anatomical differences.

The only work on spine tumor classification that caught our attention in which Park et al.[17] presented a pathology-rich dataset aligned with the latest WHO classification, including molecular biomarkers. However, its lack of imaging data and single-institution scope limit its utility for radiology-pathology integration and algorithm generalization.

3 Proposed Methodology for Spinal Tumor Classification

Our proposed methodology is as follows:

- Firstly, we segment tumors and CSF regions from 2D MRI slices based on features. Tumor-overlapping CSF regions are processed to isolate and remove tumors, generating two datasets: tumor-free spine images and extracted tumor regions, further augmented for robust model training.
- Tumor data augmentation, as outlined in Fig. 2, to ensure sufficient diversity and balance in the dataset. Starting with a sequence of 2D T2-weighted MRI slices, we reconstruct the corresponding 3D lumbar spine volume following the approach described in [8]. Post reconstruction, we apply denoising techniques to enhance image quality and anatomical detail. We then synthetically augment the dataset by inserting tumor-like structures along the cerebrospinal fluid (CSF) regions in healthy MRI volumes, imitating common tumor appearances and spatial distributions. This augmentation not only doubles the dataset size but also introduces anatomical variability crucial for generalizing the classification model.

- For classification, we employ a two-stage strategy. First, augmented and original images are passed through a CNN model trained to categorize tumor type: intramedullary, intradural extramedullary, or extradural.
- In order to improve localization-aware classification, we incorporate vertebrae labeling using the approach described in [16], enabling the model to consider spatial priors based on tumor proximity to specific vertebral levels. This classification-centric pipeline allows for precise tumor type identification, ultimately supporting clinicians in diagnosis and treatment planning with enhanced accuracy.

3.1 Segmenting Tumor and Cerebrospinal Fluid

From the 2D MRI slices containing tumors, we perform segmentation of both the tumor and the cerebrospinal fluid (CSF) regions using a Random Forest classifier [12] trained on relevant pixel-wise features. These features include raw intensity values, neighborhood statistical measures (such as local mean, standard deviation, and entropy), gradient-based descriptors, spatial coordinates to incorporate positional context, and multi-scale texture descriptors such as Local Binary Patterns (LBP) and Gabor filter responses. Once the segmentation is complete, we identify the subset of images where the tumor overlaps with the CSF region. In these cases, we isolate and remove the tumor component from the image, thereby generating corresponding spine images without tumors. We also mark the end of L5 vertebrae in each CSF using [16]. This process results in two distinct datasets: one comprising tumor-free spine images and another containing the extracted tumor regions. These datasets are then subjected to data augmentation techniques to enhance variability and support robust model training.

3.2 Tumor Data Augmentation

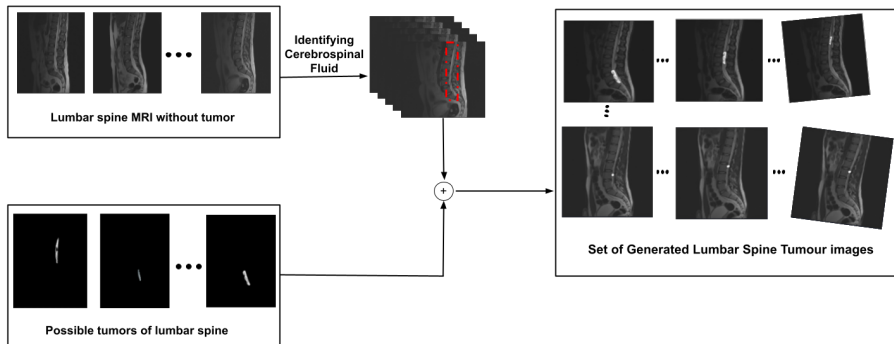


Fig. 2: Flowchart for proposed data augmentation method

Automating the process of spine tumor/lesion segmentation is challenging because there is a shortage of public data on spine tumors. So to expand the size of the dataset in the first stage, we employed a novel data augmentation process to increase the size of our dataset of lumbar spine MRI scans.

As illustrated in Fig. 2, the process begins by extracting tumors from each slice to collect various types of tumors. We then identify each slice’s cerebrospinal fluid (CSF), based on a predefined mask. For each slice, starting from T11, we add a tumor of each type and slide it one slice at a time along the CSF from T11 to L5. The tumor size in each slice is determined by the original size of the tumor in its corresponding slice. This approach allows us to obtain 2D MRI scan slices of the lumbar spine, each depicting a tumor. To account for potential slight rotation during image capture, we deliberately rotate the images by an angle between 1 and 10 degrees to the left and right. This strategy effectively doubles our dataset, providing us with an increased abundance of 3D MRI data for the lumbar spine. More precisely, suppose X , a 2D T2-weighted MRI, has a tumor x . Initially, we separate the tumor x from X and generate two images: I_x , which contains only x , and $I_{\setminus x}$, which contains X without x . In $I_{\setminus x}$, the missing data (due to removing tumor) is regenerated using bicubic interpolation [8]. Next, we extract the CSF from each image based on expert markings.

Algorithm 1 Proposed Data augmentation for Lumbar Spine T2 MRI

Input: Original dataset D of lumbar spine MRI scans with tumors

Output: Augmented dataset D' with increased number of images

begin

Step 1: For each MRI scan $X \in D$

Step 1a: Extract tumors x from each slice of X to get I_x and $I_{\setminus x}$

Step 1a: For each $I_{\setminus x}$ regenerate the missing data using
bilinear interpolation

Step 1b: Identify CSF in each slice based on the predefined mask

Step 1c: For each slice from T11 to L5

Step 1c(i): For each tumor type in I_x

Slide tumor x along CSF, starting from T11,
shifting by 1 mm

Rotate tumor x to match CSF curvature

Generate new image X' with tumor x in each
slice

Store X' in augmented dataset D'

Step 1c(ii): For each image with tumour

j=1

while j $\leq$ 10

Apply rotation to X by an angle j along left and
right directions to get rotated images X_{left}^j and

X_{right}^j

Store X_{left}^j and X_{right}^j in augmented dataset D'

Step 2: Output the augmented dataset D'

end

The tumor x is slid in each slice of the spine image that initially is devoid of tumor, along the CSF, starting from T11 and shifting by 1 mm (approximately 3 pixels) to generate another image. During this process, we account for the curvature of the CSF by rotating the tumor to match the CSF's curvature, ensuring a proper fit within the CSF. This sliding continues until we reach L5. This generates approximately 20 images from one original image. We repeat this process for each tumor type and each slice. The tumor size in each slice increases towards the center, and it decreases again as it moves towards the right, following the sagittal axis's movement from left to right during image capture for each patient's data. The algorithm of this complete augmentation process is mentioned in Algorithm 1.

No spurious or garbage data is generated, every 2D image that is generated can be used for training any deep learning model. A part of the augmented dataset for spine tumor images appears online ⁴. This is solely to test the effectiveness of the proposed model. Since we have worked with the data for sagittal MRI slices of the lumbar spine, taken with a slice gap of 3 to 5 mm, we have reconstructed the missing data using a combination of bilinear and bicubic interpolation [8] and extracted the data for consecutive intermediate slices. This process also helps in data augmentation by increasing the number of 2D slices. After data augmentation, we executed automated tumor classification as the next step of our method.

3.3 Tumour Classification and localisation

Tumor Classification: We have developed a Convolutional Neural Network (CNN) architecture with six convolutional layers, four max-pooling layers, and two fully connected dense layers to classify tumors into one of the three types of tumors shown in Fig. 1. The configuration in Fig. 3 is a setup designed to improve CNN's ability to identify minute details in medical pictures, especially those related to tumors of the lumbar spine.

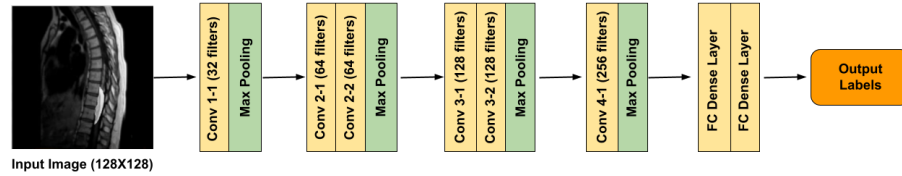


Fig. 3: Proposed CNN architecture for tumor classification

The architecture begins with an input layer with a shape of $128 \times 128 \times 3$. The first convolutional layer applies 32 filters of size 3×3 with ReLU activation and *same* padding to the input image. This is followed by a max pooling layer

⁴ [link](#)

with a pool size of 2×2 . The second and third convolutional layers each use 64 filters of size 3×3 with ReLU activation and *same* padding, followed by a second max pooling layer with a pool size of 2×2 . The fourth and fifth convolutional layers each apply 128 filters of size 3×3 with ReLU activation and *same* padding. This is followed by a third max pooling layer with a pool size of 2×2 . The sixth convolutional layer uses 256 filters of size 3×3 with ReLU activation and *same* padding, followed by a fourth max pooling layer with a pool size of 2×2 .

After the convolutional and pooling layers, the architecture includes a flattened layer, followed by two fully connected dense layers. The first and second dense layers have 256 and 128 units, respectively, with ReLU activation and a dropout rate of 0.5. The output layer consists of three units, one for each tumor type, with a softmax activation function to handle multi-class classification.

The performance of the CNN model, denoted by $P(L)$, is dependent on the number of layers L and a constant c that depends on the type and size of the dataset:

$$P(L) = \frac{1}{|L - c| + 1}$$

Through experiments and iterative refinement, we have validated the efficacy of this specific CNN arrangement for lumbar spine tumor classification. By adeptly extracting and discerning pertinent features, our modified CNN architecture demonstrates a heightened ability to classify lumbar spine tumors accurately, thereby holding significant promise in medical image analysis and diagnosis.

Tumor Localisation: To enhance localization of spinal tumors, we incorporate vertebrae labeling into the preprocessing pipeline using the method detailed in [16]. This approach performs panoptic segmentation of lumbar vertebrae, assigning unique labels to each vertebral level in the MRI slices. By doing so, we are able to encode spatial priors that provide anatomical context for tumor localization. Specifically, the model gains awareness of the tumor’s proximity to known vertebral landmarks, which is particularly useful for improving the interpretability and accuracy of classification tasks that depend on spatial positioning. This vertebrae-aware information is integrated as an additional input channel or as positional features during model training, enabling the network to better differentiate tumors based on their anatomical location along the spinal axis.

4 Results

In this section, we discuss about the results of data augmentation as well as data classification.

4.1 System Specifications and Runtime Environment

Performance analysis and a comparative study were conducted on a computing platform featuring a high-performance Intel i5 processor with a clock speed of

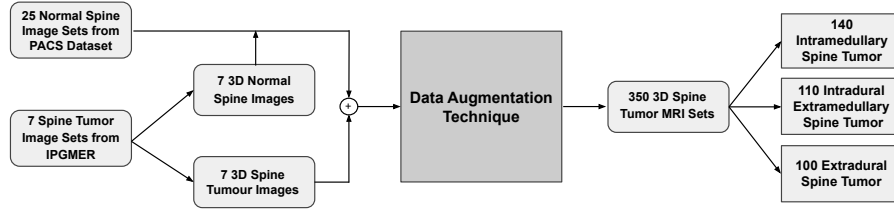


Fig. 4: Assessing the impact of our novel data augmentation method on data variability: before and after analysis

4.90 GHz and 16 gigabytes of RAM. The GPU of the NVIDIA Tesla K80 has 12 GB of VRAM, and the GPU runtime includes an Intel Xeon CPU @2.20 GHz, 13 GB of RAM, and 12 GB of GDDR5 VRAM. The results of our proposed tumor segmentation, localization, and classification methods, are presented below.

4.2 Dataset Description

With the permission of the patients and following the evaluation of the IPGMER ethics committee⁵, we obtained seven different types of spine MRI tumor data, as well as a few healthy spine images from IPGMER, Bangur Institute of Neurosciences in 2018. We also obtained some healthy spine data from the dataset [4]. We have considered only T2-weighted MRI as the tumor is most prominent in this modality [9].

4.3 Data Augmentation

We applied the proposed data augmentation technique across seven data types to generate a total of 30,000 augmented 2D spine MRI samples and 350 augmented 3D spine MRI datasets, as illustrated in Fig. 4. Among the 3D datasets, 140 correspond to Intramedullary Spine Tumor cases, 110 to Intradural Extramedullary

⁵ Memo No. - IPGME&R/IEC/2017/449

Input Images							
Tumor Type	Intramedullary	Intramedullary	Intramedullary	Intradural Extramedullary	Intradural Extramedullary	Extra Dural	Extra Dural
Origin	Spinal Cord	Spinal Cord	Spinal Cord	Nerve Roots	Nerve Roots	Surrounding bony or soft tissue structure	Surrounding bony or soft tissue structure

Fig. 5: Results for some examples of tumor classification and determination of origin

Spine Tumor cases, and 100 to Extradural Spine Tumor cases. All the mentioned generated results are checked and verified by experts.

4.4 Spine tumor classification

The results of spine tumor classification were validated by doctors and physiologist. For mathematical validation, we use performance evaluation metrics.

The following are the outcomes when the proposed tumor image data is fed into the combined model of tumor classification. type and the origin of the tumor, and also the type of the tumor as shown in Fig. 5. The accuracy for localization and classification of each of these individual types of tumor are mentioned in Table 1.

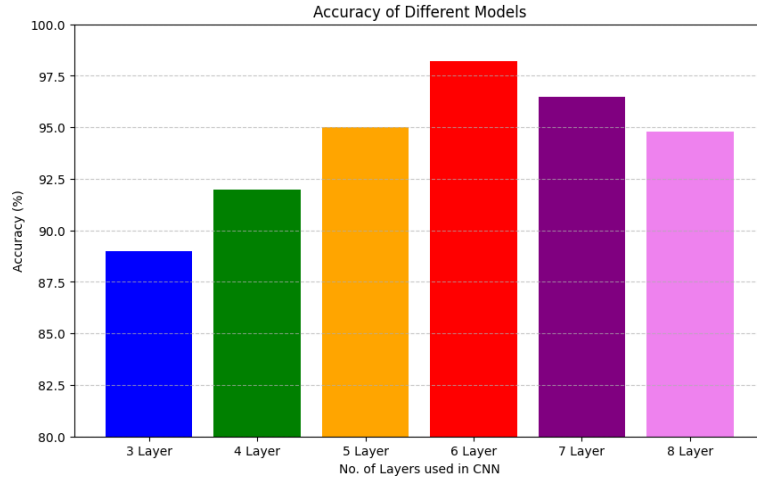


Fig. 6: Analyzing the impact of varying the layer depth of CNN on training accuracy.

The metric used for calculating accuracy is 3D multi-class accuracy [20], which evaluates the performance of a machine learning or deep learning model that deals with three-dimensional data (such as images or volumetric data) and involves classification tasks with multiple classes. It measures the frequencies of correct class label prediction by the model for the given 3D data across all classes. It can classify 3D data accurately across multiple categories, and it is calculated as follows:

$$Class\ Accuracy = \frac{TN + TP}{TN + TP + FP + FN} \quad (1)$$

Ablation Study for classification of tumor: While proposing the CNN, we evaluated different configurations of CNN layers and observed a significant

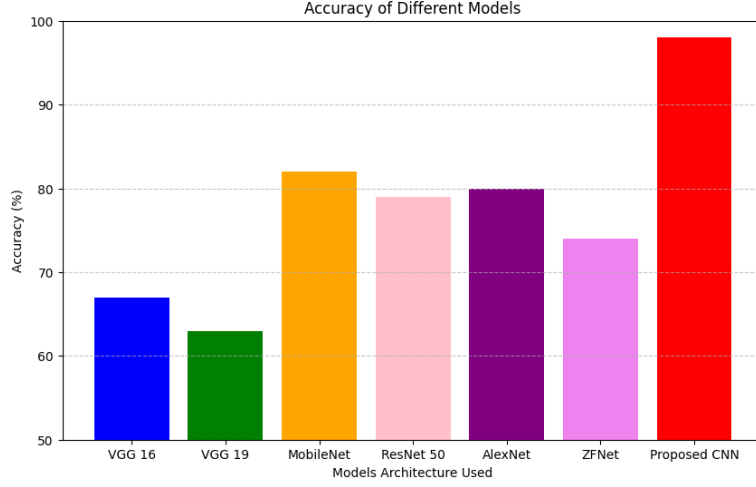


Fig. 7: Comparison of accuracy among various CNN architectures.

Table 1: Localisation and Classification Accuracy for different types of tumor

Tumor Type	Accuracy	Dice Score
Intramedullary	99.5%	0.98
Intradural Extramedullary	98.6%	0.97
Extra Dural	99.1%	0.98

trend in the results. As depicted in Fig. 6, a CNN architecture consisting of six layers yielded the most promising outcomes among all the variations we explored.

Comparison and Analysis of classification methods: Our model demonstrates proficiency in classifying three types of tumors: Intramedullary, Intradural-Extramedullary, and Extra-Dural Tumors. Through testing and comparing against established model architectures such as VGG16 [19], VGG19 [19], MobileNetV2 [18], ResNet50 [13], AlexNet [14], and ZFNet [23], our proposed CNN architecture has consistently outperformed them all when tested on both real data and synthetic data. This superiority is evident in the model accuracy results depicted in Fig. 7.

This finding underscores the critical role of model architecture in achieving optimal performance in our tumor classification task. By leveraging a six-layer CNN configuration, our model demonstrates enhanced capabilities for effectively discerning between different types of tumors with higher accuracy and reliability. This insight highlights the importance of fine-tuning the architectural aspects of neural networks to achieve the best possible outcomes in medical image analysis tasks like tumor classification.

We observe a gradual and consistent decrease in loss values across 40 training epochs of training and validation, as shown in Fig. 8. This downward trend signifies a crucial aspect of the efficacy and robustness of our model. The steady

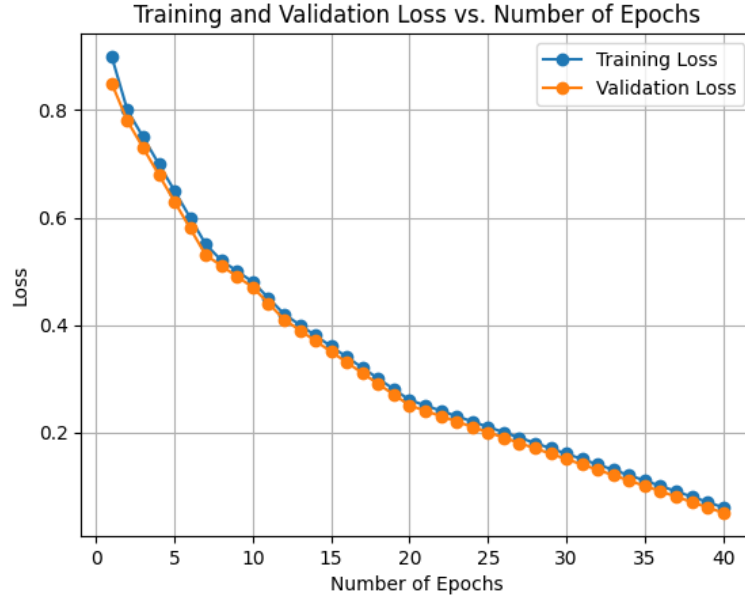


Fig. 8: Loss curve during training & validation.

decline in loss values throughout multiple epochs shows how well our model generalizes to previously unseen validation data and efficiently learns from the training set. This phenomenon is important for machine learning, especially for tasks where precise predictions are critical for clinical diagnosis and decision-making, such as tumor categorization.

The training and validation loss curves both show a downward trend, suggesting that our model can detect complex patterns in the training data and can prevent overfitting by performing well on unknown validation data. A well-designed model is shown by this harmony between generalizing to new data and learning from the training set. We have generated a confusion matrix to visually represent the classification performance of our model, as illustrated in Fig. 9.

5 Conclusion and Future Works

By leveraging deep learning and advanced imaging techniques, notable advancements have been achieved in classifying spinal cord tumors. However, progress is still hindered by the limited availability and variability of spine tumor datasets, which can introduce bias and affect generalization performance. While models such as the one developed by Zhuo et al. [26] have demonstrated impressive training accuracy (up to 96%), they face challenges in differentiating complex tissue structures due to insufficient and non-diverse data.

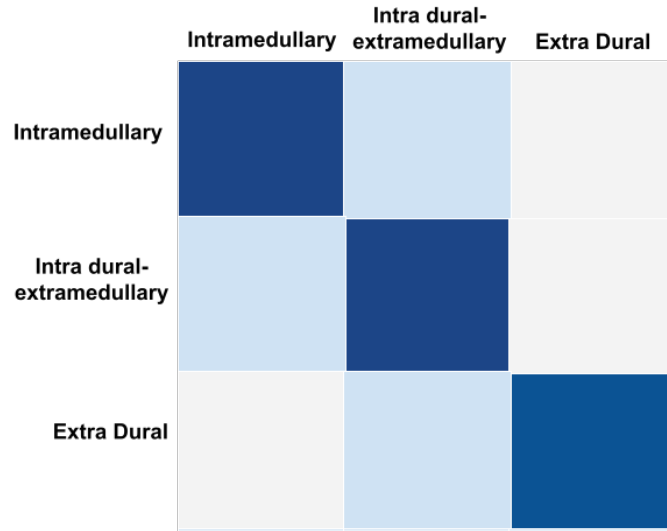


Fig. 9: Visualizing performance of identification of tumor type: Confusion matrix analysis of our CNN model.

What distinguishes our approach is the exceptional quality and diversity of training data, carefully curated through real-time data integration and extensive collaboration with medical professionals across varied clinical environments. Our method achieves a classification accuracy of 98% using a 6-layer convolutional neural network (CNN), efficiently capturing discriminative tumor features across different imaging conditions.

Furthermore, we introduce a robust data generation pipeline that automates the creation of high-fidelity 3D spinal cord tumor datasets. This includes a novel data augmentation strategy that addresses the scarcity of annotated data by synthesizing realistic and diverse tumor variations. The generated datasets enhance classification accuracy and improve model robustness in real-world clinical settings. In the future, we aim to implement other image processing tasks, such as spine tumor segmentation, etc., with the generated dataset.

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